

# DeepSeek- V2

A Strong, Economical, and Efficient  
MoE Language Model

Combines Multi-Head Latent Attention (MLA) with  
DeepSeekMoE architecture for state-of-the-art  
performance at dramatically reduced inference cost.

## KEY METRICS

**236B**

Total Parameters

**21B**

Activated per Token

**128K**

Context Length

# The LLM Scaling Dilemma

## ! The Challenge

### 1 Dense Scaling Cost

Dense models scale poorly —  $2\times$  params  $\approx 2\times$  compute and memory cost at every layer.

### 2 KV Cache Bottleneck

KV cache grows linearly with context length, bottlenecking inference at long contexts.

### 3 Memory Wall (MHA)

Multi-Head Attention stores full K/V per head — hitting a memory wall at scale.

vs

## ✓ DeepSeek-V2 Approach

### → MoE Sparse Activation

Activate only 21B of 236B params per token — full model capacity at fractional cost.

### → MLA: 93.3% Cache Reduction

Compress KV to compact latent vector — 93.3% KV cache reduction vs. standard MHA.

### → 5.76× Throughput Gain

5.76× generation throughput vs. DeepSeek 67B dense baseline at comparable quality.

# DeepSeek-V2 Architecture Overview

## MODEL SPECS

**236B** total parameters (MoE architecture)

**21B** activated per token

**128K** token context window

**8.1T** tokens in pretraining corpus

## TWO CORE INNOVATIONS

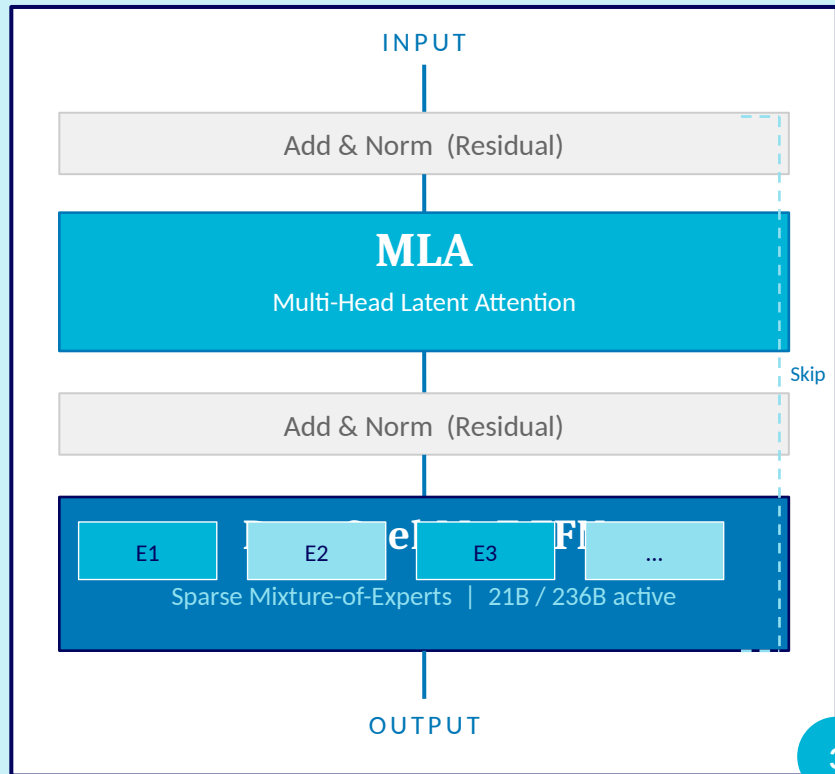
**MLA** Multi-Head Latent Attention

Replaces standard MHA — compresses KV to latent vectors

**DeepSeekMoE** Mixture-of-Experts FFN

Replaces dense FFN layers — sparse activation at scale

## Transformer Block



# Multi-Head Latent Attention (MLA): 93.3% KV Cache Reduction

## THE PROBLEM

Standard MHA caches full K,V per head per token — memory grows  $O(n \cdot h \cdot d)$  with sequence length, creating an inference bottleneck.

## MLA SOLUTION

Low-rank joint compression: cache only compact latent vector  $c^{tko}$ . Reconstruct K, V, and decoupled RoPE positional key on-the-fly via learned projection matrices.

# 93.3%

KV cache reduction

## ATTENTION METHOD COMPARISON

| Method | Cache Size                        | Performance          |
|--------|-----------------------------------|----------------------|
| MHA    | Full K,V per head                 | Baseline             |
| GQA    | Shared K,V groups                 | Slightly lower       |
| MQA    | Single K,V                        | Lower                |
| MLA    | Latent vector only<br>6.7% of MHA | Better than MHA<br>✓ |

Source: DeepSeek-V2 paper (arxiv 2405.04434)

# DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

## SHARED EXPERTS — Always Active ( $N_s$ )

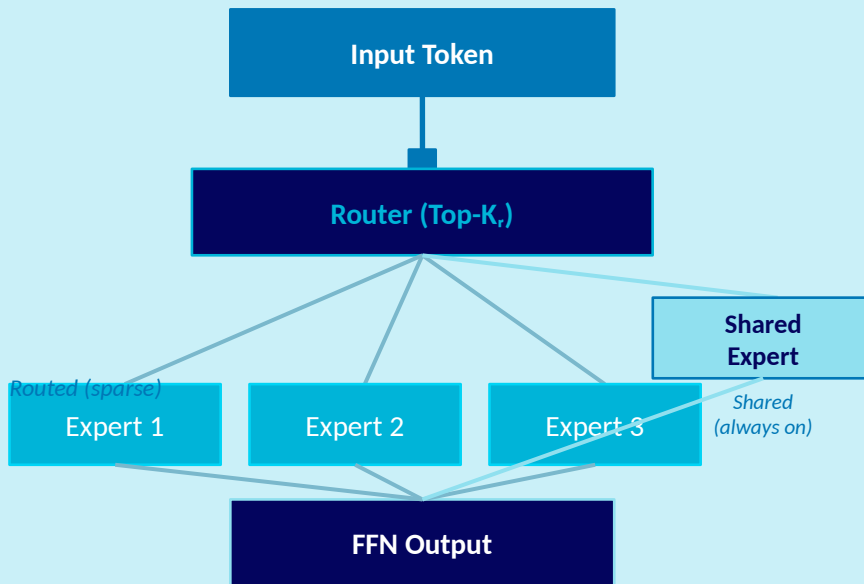
Always activated — capture universal knowledge across all inputs. Guarantee consistent base representations regardless of routing decisions.

## ROUTED EXPERTS — Sparse Activation ( $N_r$ , Top-K<sub>r</sub>)

Activated selectively via Top-K routing — enable token-level specialization. Fine-grained segmentation maximizes expert utilization.

### KEY DESIGN FEATURES

- Fine-grained segmentation → higher specialization
- Device-limited routing → reduced cross-node traffic
- Auxiliary load-balancing losses → balanced utilization



MoE Routing: Each token processed by shared + top-K routed experts

# 42.5% Training Cost Reduction and 5.76× Throughput vs. DeepSeek 67B

## DeepSeek 67B

Dense Baseline

Dense MoE

67B parameters

VS

## DeepSeek-V2

MoE + MLA

MLA + DeepSeekMoE

236B total / 21B active

Training Cost

Baseline



**42.5%**  
lower cost

✓ Improvement

KV Cache Size

100% (full)



**93.3%**  
reduction

✓ Improvement

Max Throughput

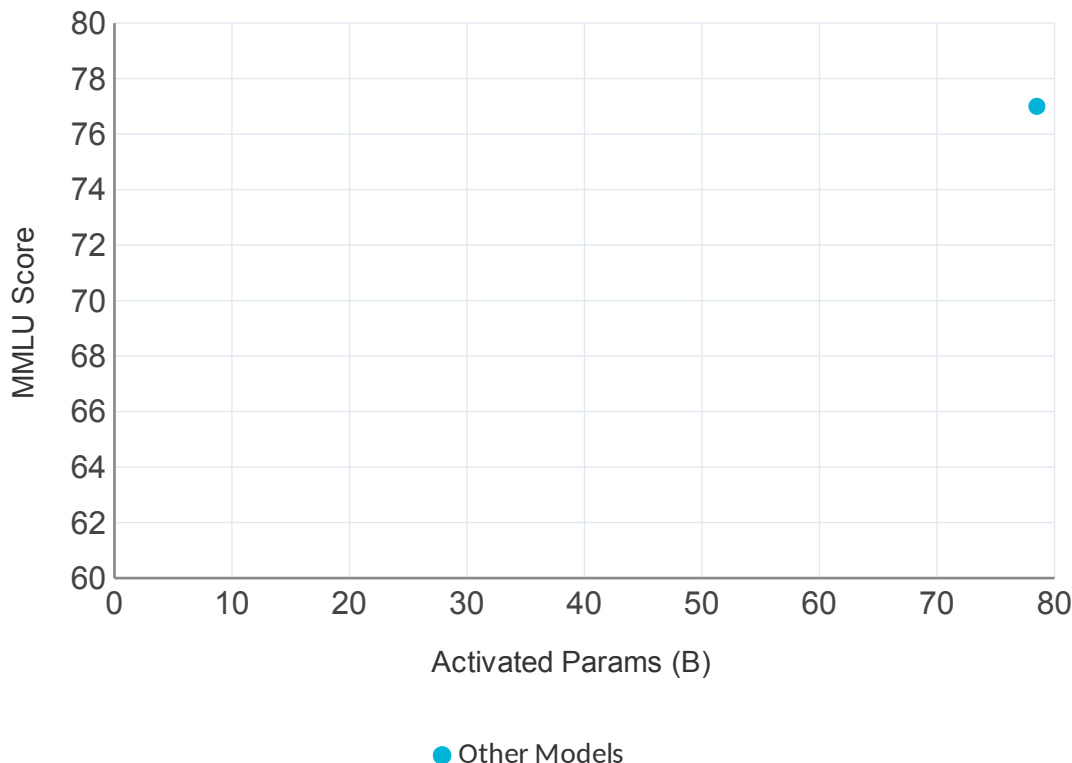
1×



**5.76×**  
higher throughput

✓ Improvement

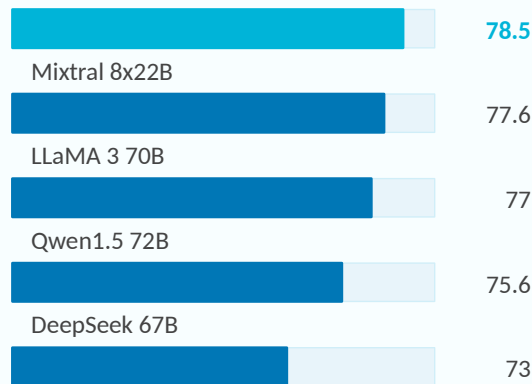
# DeepSeek-V2 Tops MMLU at Only 21B Activated Parameters



## Pareto-Optimal Position

DeepSeek-V2 achieves 78.5 MMLU with only 21B activated params — outperforming 70B-class dense models.

### DeepSeek-V2



Source: arxiv 2405.04434

# DeepSeek-V2 Chat (RL): Alignment Results

## Alignment Pipeline

### Pretraining

8.1T tokens · high-quality multi-source corpus



### Supervised Fine-Tuning (SFT)

1.5M conversational sessions



### RL via GRPO

Group Relative Policy Optimization

# 38.9%

AlpacaEval 2.0 LC Win Rate

LC Win Rate

# 8.97

MT-Bench Overall Score

Multi-turn

# 7.91

AlignBench (Chinese)

★ Best Open-Source



# Key Takeaways

0

1

MLA reduces KV cache by 93.3% vs MHA while improving performance — strictly superior to GQA/MQA

0

4

5.76× generation throughput improvement makes DeepSeek-V2 practical for large-scale deployment

0

2

DeepSeekMoE trains the 236B-param model at 42.5% lower cost than a 67B dense baseline

0

5

DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source in English

0

3

Only 21B activated parameters match or exceed 70B-class dense models on MMLU (78.5 vs 77.0)